Burugapalli Rohan

Computer Science Student

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Professional Summary

Computer Science student specialising in NLP, machine learning, and AI application development. Experienced in fine-tuning transformer models, building RAG pipelines, and deploying LLM-based systems using Python, PyTorch, and modern AI frameworks. Skilled in developing end-to-end solutions from model training to production deployment across AI, and data science.

Technical Skills

Programming: Python, C++, Java
ML/AI: PyTorch, TensorFlow, HuggingFace Transformers, NLP, RAG, LLMs, Prompt Engineering
Data: Pandas, NumPy, FastEmbed, Vector Databases, SQL
Frameworks/Libraries: FastAPI, LlamaIndex, LangChain, Groq, REST APIs, Git
Relevant Coursework: DSA, OOPs, DBMS, Data Science, Cryptography, Blockchain, LLMs
Languages: English, Telugu, and Hindi

Projects

MCP Tool-Response Security Firewall

- Built an ML-based security layer for Model Context Protocol (MCP) pipelines detecting prompt injection and tool-poisoning attacks using NLP classifiers.
- Trained and deployed text classification models to inspect LLM tool descriptions and responses, flagging adversarial inputs before execution.
- Demonstrated end-to-end AI security pipeline integrating threat detection with agentic LLM workflows using Python and HuggingFace Transformers.

PDF and Web Content Study Assistant

- Developed full-stack RAG application with FastAPI and LlamaIndex enabling PDF upload and URL scraping for intelligent Q&A and automated summarization of educational content.
- Engineered high-performance retrieval pipeline using Groq LLM inference and FastEmbed vector embeddings to deliver context-aware answers with <2s response times.
- Built responsive 4-view interface (Upload, Web Import, Chat, Summary) with streaming responses using vanilla JavaScript for real-time user interaction.

AI News Aggregator

- Built a generative AI pipeline using Groq (Llama 3.3 70B) to scrape Anthropic and OpenAI RSS feeds and YouTube channels, generating concise summaries and ranking content against a personalized user interest profile.
- Implemented duplicate tracking via Neon Postgres and automated daily delivery via Gmail SMTP, with fully configurable scheduling through GitHub Actions cron workflows.
- Deployed on a fully free production stack with support for multi-provider LLM fallback (Groq, Gemini, OpenAI) and user-customizable interest profiles, YouTube sources, and email digest size.

AES and RSA Data Storage

- Designed and implemented a two-layer envelope encryption scheme ensuring data confidentiality and integrity.
- Achieved AES encryption/decryption throughput of 200 MB/s on multi-megabyte datasets.
- Maintained RSA key encryption/decryption latency under 10 milliseconds per key, optimizing performance for multiple authorized roles.

ZK-Based Private Token Transfer System

- Engineered privacy layer for Ethereum using Circom zk-SNARK circuits and Groth16 proving system, hiding sender, receiver, and amount while maintaining trustless on-chain verification.
- Designed shielded pool architecture with Merkle tree commitment scheme, nullifier-based double-spend prevention, and Poseidon hash for ZK-friendly cryptographic operations.
- Implemented end-to-end system: Solidity contracts with incremental Merkle updates, JavaScript proof generation SDK, and web interface for 3-operation flow (shield, transfer, unshield).

Phishing Email and URL Detector

- Developed an AI-powered phishing detection system integrating Natural Language Processing and machine learning for email and URL threat analysis.
- Fine-tuned DistilBERT transformer for phishing email classification achieving 92% accuracy and 500 ms inference time using PyTorch and HuggingFace.
- Built a hybrid risk scoring pipeline combining NLP email semantics and URL-based feature extraction using scikit-learn for multi-signal detection.

Work Experience

Internship – Vector Techniques (BLDC Motors) June – July 2023
Embedded Systems Intern Hyderabad, Telangana

- Gained corporate world experience focusing on embedded C programming during internship at Vector Technics.
- Analyzed performance data to identify and resolve real-time system inefficiencies.
- Developed practical skills in embedded systems and real-time programming applications.

Internship – Everest DG & EIT Global June – July 2024
Design Intern Dubai, UAE

- Worked on web development using WIX, UI/UX design with Canva, and digital marketing content creation.
- Used Figma to design clean website layouts and app screens, creating interactive prototypes to help the team visualize project ideas.
- Contributed to multiple client-facing projects involving emerging technologies.

Research

Thesis: Zero Trust Enhanced Defense Mechanisms for Mitigating Social Engineering in Online Social Networks
BITS Pilani, Dubai Campus February 2026 – May 2026

- Designed a novel dual-encoder Zero Trust pipeline extending NIST SP 800-207 to social engineering detection on LinkedIn; Stage 1 uses SecureBERT for message-level manipulation detection and Stage 2 uses ModernBERT for full conversation behavioural sequence modelling.
- Introduced conflict-aware MLP fusion computing cosine dissimilarity between encoder embeddings as an explicit SE signal, reducing FPR from 45.6% to 23.7% and achieving F1=0.833 and AUROC=0.861 on SEConvo, outperforming ConvoSentinel (F1=0.74) and GPT-4-Turbo (F1=0.67).
- Trained on SEConvo and SE-VSim totalling 490 LinkedIn-style conversations; implemented LIME, KernelSHAP, and trust score distribution analysis for Zero Trust auditability.

Education

Birla Institute of Technology and Science Dubai, UAE
B.Tech in Computer Science and Engineering September 2022 – May 2026